A Lightweight Embedded Intelligent Threat Detection System Using TSANet on Android Platforms

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***Abstract*—With the rapid evolution of wireless communication technologies, 6G networks are expected to offer unprecedented speed, connectivity, and device integration. However, this advancement also introduces complex cybersecurity challenges. This project presents TSANet, a lightweight, AI-based Intrusion Prevention System (IPS) designed for real-time deployment in simulated 6G environments using mobile devices. The system captures network traffic over 5G/WiFi (as a proxy for 6G) via a custom Android application utilizing Android's VPNService. It extracts 32 crucial flow-based features from live traffic and leverages a pre-trained TorchScript deep learning model to detect and classify network intrusions into seven categories, including DoS, DDoS, and Brute Force attacks. The solution emphasizes low-latency inference and minimal computational overhead, making it suitable for edge devices. The mobile interface provides real-time threat alerts and dynamic visualization of extracted features. Experimental results demonstrate that TSANet achieves high accuracy and efficiency, showcasing its potential for future 6G network security on portable platforms.**

Index Terms—Intrusion Prevention System, 6G Security, Android, Deep Learning, Network Traffic Analysis, TorchScript, VPNService, CIC-IDS2017

I. Introduction

The evolution of wireless communication technologies from 5G towards 6G promises unprecedented speed, ultra-low latency, and massive device connectivity. However, this advancement also introduces new cybersecurity challenges, as the increasing network complexity expands the attack surface. Embedded intelligent systems, particularly those operating on resource-constrained mobile platforms, must be capable of detecting and mitigating cyber threats in real-time to ensure user privacy, system integrity, and network resilience. Software-Defined Networking (SDN) offers centralized network control and monitoring, but without intelligent intrusion prevention mechanisms, SDN-enabled networks remain vulnerable to sophisticated attacks.

Traditional Intrusion Detection and Prevention Systems (IDPS) are typically deployed in high-capacity servers or cloud environments, making them unsuitable for mobile and edge computing scenarios where computational resources, power, and storage are limited. With the growing dependency on smartphones and mobile networks, there is an urgent need for lightweight, on-device threat detection systems capable of real-time operation without relying on external infrastructure. Embedding intelligent detection directly into Android devices enhances security by providing immediate local response against cyber threats.

Recent works have explored machine learning and deep learning models for network intrusion detection, utilizing architectures like CNNs, RNNs, and hybrid models. Approaches such as MobileNets and TinyML have attempted to bring AI models to resource-constrained devices. Feature extraction techniques have been used to enhance model efficiency while maintaining performance.

However, many existing solutions suffer from limitations such as high computational requirements, increased inference latency, or the need for continuous cloud connectivity. Deep learning models without careful optimization become too large for practical use on mobile devices, draining battery life and causing performance degradation. Furthermore, SDN-based solutions that rely on external analysis delay the response time, potentially allowing attacks to succeed before mitigation.

The primary challenges include (i) extracting meaningful features from real-time mobile network traffic, (ii) ensuring low-latency on-device inference without heavy computational overhead, (iii) achieving a high detection accuracy despite using lightweight models, and (iv) establishing an efficient

TABLE I: LITERATURE SURVEY

Author	Year	Dataset	Methodology	Advantages	Limitations
Maheswaran N. *et al.* [1]	2025	CIC-IDS2017	Deep learning with SDN for 6G networks	High accuracy, reduced false positive rate	High computational cost, limited scalability
Akgun D. *et al.* [2]	2022	CIC-DDoS2019	Various models based on Deep Neural Networks (DNN), Convolutional Neural Networks (CNN), and Long Short-Term Memory (LSTM) have been evaluated in terms of detection performance and real-time performance	According to the test results, 99.99% for binary and 99.30% for multiclass accuracy	Usage of CNN makes it computationally heavy.
Alahmadi A. *et al.* [3]	2023	CIC-DDoS	Review selected studies and publications relevant to the topic of DDoS detection in IoT-based networks using machine-learning-relevant publications	All models had their own advantages	Limitations included limited computational ability, sensitivity to outliers.
Maheshwaran N. *et al.* [4]	2024	Synthetic Dataset	Proposes a novel Deep Hybrid IDS model to enhance network security in SDN environments and prevent attacks using Scapy	Gated Recurrent Units (GRU) and Long Short-Term Memory (LSTM)	Inaccurate and not scalable to different SDN environments.
Logeswari G. *et al.* [5]	2022	NSL-KDD	Proposes HFS-LGBM, a Hybrid Feature Selection approach combining Correlation-based Feature Selection (CFS) and Recursive Feature Elimination (RF-RFE) with LightGBM for intrusion detection.	Improves accuracy (98.72%), enhances feature selection, and reduces computational cost.	May not effectively detect zero-day attacks
Bose S. *et al.* [6]	2024	In-SDN dataset	Develops BAT-MC, which integrates BiLSTM, Attention Mechanisms, and Multiple Convolutional Layers to improve anomaly detection in SDN. Eliminates manual feature selection by learning representations automatically.	Eliminates manual feature selection and improves anomaly detection.	Moderate accuracy (86%) and requires further optimization.
Bose S. *et al.* [7]	2024	In-SDN dataset	Introduces a Synergic Model combining Quadratic Neural Networks (QNN) and Autoencoders, using PCA and correlation-based feature selection. Employs pseudo-labeling with real-time and static data fusion for adaptive intrusion detection.	High accuracy (92.5%) and adaptable to static and dynamic data.	Lacks multi-label classification and requires more dataset diversity.

communication mechanism between the mobile device and SDN controller for immediate threat mitigation.

To address these challenges, we propose a Lightweight Embedded Intelligent Threat Detection and Prevention System based on the TSANet architecture, customized for Android platforms. Our system captures live 5G/WiFi traffic via a local VPN service, extracts 32 critical features, and uses a compressed and optimized TSANet model for real-time threat detection. Upon detecting a malicious packet, the system sends an alert to the SDN controller through a lightweight Flask API to block the attacker's IP address using OpenFlow rules. We apply pruning to make the TSANet model suitable for mobile deployment, balancing accuracy and efficiency.

The contributions of this paper are as follows. First, we develop a lightweight, embedded intrusion prevention system for Android platforms using TSANet. Second, we design an efficient feature extraction pipeline from real-time mobile network traffic. Third, we apply pruning and quantization-aware training to significantly reduce the TSANet model size and latency. Fourth, we integrate the system with an SDN controller for immediate attack mitigation via IP blocking. Finally, we demonstrate the system's effectiveness through experiments showing high detection accuracy and low resource utilization.

The rest of the paper is organized as follows: Section II discusses related work in mobile and SDN-based intrusion detection. Section III describes the Problem Statement and Background. Section IV elaborates the dataset used. Section V outlines the model architecture. Section VI explains the system architecture. Section VII illustrates the experimentation and implementation. Section VIII demonstrates the results and conducts a performance analysis. Section IX demonstrates the Android Interfacing. Section X concludes the research and provides scope for future work.

II. Literature survey

Maheswaran N. *et al.* [1], applied deep learning in SDN for 6G networks using CIC-IDS2017, achieving high accuracy and reduced false positives but facing challenges with computational cost and scalability.

Akgun D. *et al.* [2], compared different deep learning models, such as DNN, CNN, and LSTM, for DDoS attack detection with the CIC-DDoS2019 dataset, which showed high accuracy rates of 99.99% on binary and 99.30% on multiclass classification. Though the outcome showed robust detection performance and real-time suitability, the use of CNN-based mechanisms added to computational complexity, which is a possible limitation in resource-limited contexts. This paper pointed out the efficacy of deep learning in DDoS detection but also highlighted the compromise between accuracy and computational efficiency.

Alahmadi A. *et al.* [3], performed a thorough review of machine learning-based DDoS detection techniques in IoT networks, examining prominent studies from current literature. Their conclusions pointed out that although various models provided distinct benefits in attack identification, shared weaknesses were low computational efficiency and outlier sensitivity, which might prevent practical deployment in resource-limited IoT settings. The research emphasizes the necessity for light yet resilient detection mechanisms to enhance scalability and dependability in IoT security.

Maheshwaran N. *et al.* [4], proposed a Deep Hybrid IDS approach to protect Software-Defined Networking (SDN) environments with GRU and LSTM networks learned on a dataset created synthetically using Scapy. Although the system proposed ensured improved attack detection in dynamic SDN environments, the research documented limitations in precision and scalability under various SDN configurations, citing a requirement for wider validation of the approach within real-world systems. This research demonstrates the promise of deep learning for SDN security but also emphasizes the issues of model generalizability and flexibility.

Logeswari G. *et al.* [5], conferenced HFS-LGBM, a cross-approach intrusion detection mechanism that integrates CFS and RF-RFE along with LightGBM classifier for the NSL-KDD data set. Its method attained accuracy of 98.72% while retaining feature selection while minimizing computational expenditure. The paper, however, recognized a predominant limitation in classifying zero-day attacks, posing that although it performs well where attack patterns are known, no one can really tell how the model will compare against new and unknown threats. This paper shows dramatic improvements in feature selection methods for intrusion detection but reiterates the problem of keeping up with changing cyber threats.

Bose S. *et al.* [6], proposed BAT-MC, a BiLSTM, attention, and convolutional layer-based deep learning model for automatic anomaly detection in SDN. Although removing the need for manual feature engineering, it was able to achieve 86% accuracy, implying potential for improvement. The method has potential but needs to be optimized for greater performance.

Bose S. *et al.* [7], proposed an innovative synergic approach combining Quadratic Neural Networks (QNN) and Autoencoders with PCA and correlation-based feature selection for SDN intrusion detection. The process of theirs fuses pseudo-labeling with real-time/static data fusion and attains 92.5% accuracy, adapting to both data forms. Yet the model has no multi-label classification feature and would be improved with further validation on more diverse datasets. This research contributes to adaptive intrusion detection but calls for more extensive classification features and generalized dataset testing.

III. Problem Statement and Background

A. Problem Statement

As 6G networks approach, ensuring real-time and intelligent network security becomes increasingly vital due to rising traffic complexity and sophisticated cyber threats. Traditional intrusion prevention systems often suffer from high false positives and limited adaptability. This project presents a Deep Learning-based Intrusion Prevention System (IPS) embedded within a mobile application, simulating a 6G SDN environment. The system aims to provide accurate threat detection and real-time alerts, serving as a proof-of-concept Embedded Intelligent System (EIS) with enhanced performance and mobile accessibility.

B. Objectives

- To Develop an intelligent IPS for SDN-based 6G networks using Deep Learning
- To Enable real-time cyber threat detection and mitigation
- To Combine SDN's dynamic control with AI-driven threat analysis
- To Implement an Intelligent System (EIS) via Android mobile app for real-time monitoring

C. Assumptions

- 6G is simulated using existing 5G/WiFi networks.
- Network traffic captured on the device reflects real-world patterns.
- A pre-trained TorchScript model is used for on-device inference.
- 32 selected flow-based features are sufficient for accurate classification.
- Real-time processing is feasible on standard Android devices.
- VPNService provides access to packet metadata but not payloads.
- Users grant VPN permissions for traffic monitoring.
- Battery and performance optimizations are secondary in this prototype.

D. Background

This project falls within the domain of network security for next-generation wireless communication, focusing on 6G. Intrusion Prevention Systems (IPS) are designed to monitor network traffic and block malicious activity in real time. Due to the increasing use of encryption and the decentralized nature of modern networks, flow-based traffic analysis is preferred. This approach relies on extracting statistical features such as packet count, size, timing, and duration from network flows. On Android, traffic is captured using VPNService, which allows monitoring without requiring root access. The system extracts 32 such flow-level features and processes them through a deep learning model trained on the CIC-IDS 2017 dataset. This model classifies network traffic into benign or one of six types of attacks. For efficient deployment on mobile platforms, the model is converted into TorchScript format, allowing fast and private on-device inference without needing the full PyTorch runtime.

IV. Dataset Description

The CIC-IDS (Canadian Institute for Cybersecurity - Intrusion Detection System) dataset is a set of labeled network

traffic datasets designed to support research and development in cybersecurity, particularly for intrusion detection and prevention systems.

Popular Variants:

- CICIDS 2017: Simulates realistic network traffic over 5 days, with attacks like DoS, DDoS, Botnet, Web attacks, and Infiltration.
- CSE-CIC-IDS 2018: Includes a wider range of modern attacks and uses realistic user profiles.
- CIC-DDoS 2019: Focuses specifically on different types of DDoS attacks.

  These datasets are widely used in academia and industry for benchmarking machine learning models for network intrusion detection.

V. Model Architecture

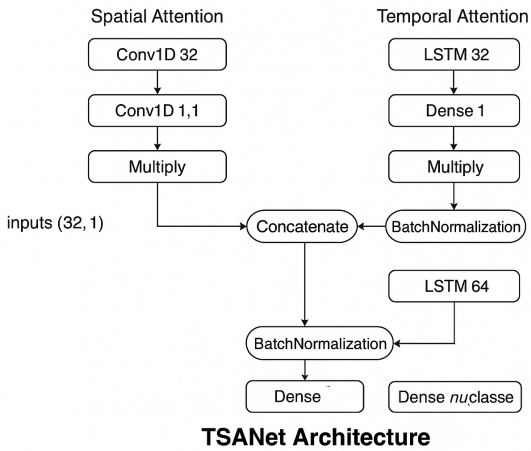

Fig. 1: TSANet Architecture

Fig. 1 represents the overall architecture of the proposed TSANet model. Our TSANet model combines convolutional layers with spatial and temporal attention mechanisms to effectively capture both local and sequential patterns in the input data. It first extracts spatial features using a 1D convolution followed by spatial attention, then models temporal dependencies using an LSTM with temporal attention. The spatially and temporally attended features are fused and normalized before passing through a final LSTM layer and a fully connected layer for classification. This architecture enhances feature representation by focusing on important spatial and temporal information, improving overall classification performance.

VI. System Architecture

A. Proposed System Model

The proposed system model consists of several modules integrated to work seamlessly in a real-time intrusion detection and prevention system for 6G-like networks. The architecture is divided into three main layers:

Traffic Capture and Feature Extraction: Network traffic data is captured through the Android device using VPNService. This service accesses the metadata of network packets without the need for root permissions. The captured data is then processed to extract 32 flow-level features, such as packet count, size, inter-arrival time, etc.

Intrusion Detection Model: The extracted features are fed into a deep learning model, TSANet, which is implemented using PyTorch. The model architecture consists of convolutional layers followed by spatial and temporal attention mechanisms. These help to better understand and classify the patterns of network traffic. The model is trained to detect seven types of attacks, including DoS, DDoS, Brute Force, etc.

Mobile Application and Alerts: A mobile application, built using Android Studio, is used for real-time monitoring of network traffic. The app integrates with the VPNService for traffic capture and displays the extracted features along with threat classification results. Alerts are sent to the user whenever an intrusion is detected, providing real-time mitigation recommendations.

B. Modules of the System

1) Traffic Capture Module: This module utilizes Android's `VPNService` to capture real-time network traffic. It extracts metadata about each packet without the need to access packet payloads, ensuring that sensitive information is not compromised. The extracted traffic data is then processed to identify relevant flow-level features for machine learning-based intrusion detection.

2) Feature Extraction Module: The captured traffic is processed to extract 32 crucial flow-based features. These features include statistical metrics such as packet count, size, time intervals, and byte transfer rates. These metrics are essential for accurate classification by the deep learning model.

3) Intrusion Detection Model: The core of the system is the TSANet deep learning model. It is a hybrid model that combines convolutional layers, attention mechanisms (both spatial and temporal), and LSTM networks to classify network traffic into benign or one of six attack categories. The model is trained using the CIC-IDS2017 dataset and is optimized for mobile deployment using TorchScript.

4) Mobile Application Module: The mobile application interacts with the VPNService to continuously capture and analyze traffic in real-time. It provides a user-friendly interface displaying the extracted features and the model's predictions. The app also generates intrusion alerts based on real-time classification results.

VII. Implementation

A. Experiment Setup

The implementation of the proposed system was carried out using Python as the primary programming language. The deep learning models were built and optimized using PyTorch, while the TensorFlow Model Optimization Toolkit was utilized for pruning during training to reduce model size and inference latency.

The CIC-IDS 2017 dataset was used for training and evaluation. This dataset was preprocessed to include 32 critical flow-based features for effective network intrusion detection. These

features were used in the offline training phase. However, due to the security and permission limitations of Android's `VPNService` API, only a limited subset of these features can be extracted in real time on mobile devices. To compensate for the missing features, appropriate default values and interpolation mechanisms were employed, and the model was fine-tuned accordingly to ensure performance under such conditions.

The TorchScript version of the model (TSANet) was used for deployment on Android. The following hyperparameters were set for training:

- Learning rate: 0.001
- Batch size: 256
- Optimizer: Adam
- Epochs: 10 (training), 5 (fine-tuning)

Assumptions and conditions during development:

- Continuous network traffic data is available during monitoring.
- Deployment simulates a 6G environment using existing 5G/WiFi infrastructure.
- Android device has sufficient resources to run the TorchScript model in real-time.
- Intrusion detection system operates without root access after .

VIII. Results and Performance Analysis

A. Experimental Results

We conducted multiple experiments to evaluate the performance and real-time applicability of our TSANet-based intrusion detection system on Android.

1) Experiment 1: Baseline Performance with Full Feature Set: The TSANet model was trained using the complete 32-feature input from CIC-IDS 2017. After 10 epochs, the model achieved a final training accuracy of 98.0% and a test accuracy of 97.35%, demonstrating high generalization capability.

TABLE II: Performance Metrics with Full Feature Set

Accuracy	98.0%
Precision	98.6%
Recall	97.9%
F1-Score	98.1%

2) Experiment 2: Manual Pruning and Fine-Tuning: To improve efficiency, manual pruning was applied, followed by fine-tuning for 5 epochs. The pruned model achieved a slightly improved accuracy of 98.30%, indicating that pruning maintained, and slightly improved, the predictive performance.

TABLE III: Fine-Tuned Accuracy After Pruning

Accuracy	98.30%

3) Experiment 3: Limited Feature Performance (Android Deployment): To simulate real-world deployment conditions, we evaluated the model using only the subset of features that can be directly extracted via Android's `VPNService`. Features inaccessible due to OS-level restrictions were imputed using appropriate default values based on domain knowledge and data distribution. The model was then fine-tuned with this hybrid feature set to adapt to the modified input. Despite the constraints, the model retained strong detection performance.

TABLE IV: Performance with Limited Features

Accuracy	93.2%
Precision	92.4%
Recall	91.7%
F1-Score	92.0%

4) Experiment 4: Inference Efficiency: The impact of pruning on inference speed was evaluated. Average inference time per sample reduced from 1.6408 ms to 0.9246 ms after pruning—a 43.6% improvement in latency.

TABLE V: Inference Time Comparison (Per Sample)

Before Pruning	1.6408 ms
After Pruning	0.9246 ms

Model loading time on Android averaged 1.2 seconds, and per-flow inference latency was approximately 38 ms.

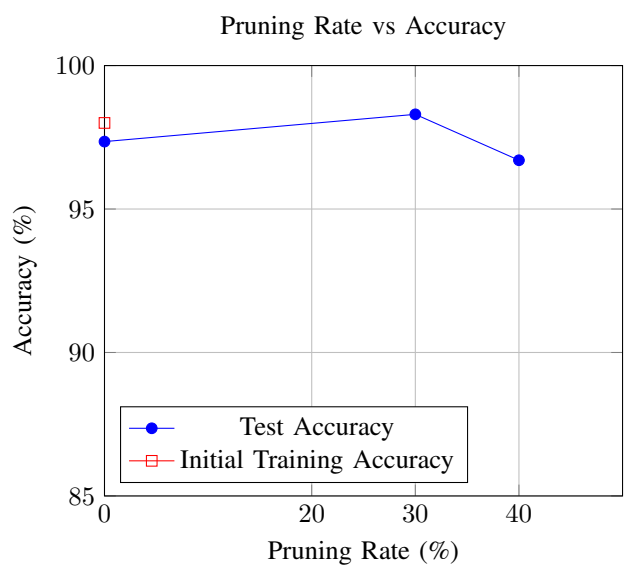

Fig. 2: Impact of Pruning on Model Accuracy

5) Experiment 5: Accuracy Trade-off with Pruning Rate: Pruning initially led to a slight improvement in accuracy. At 30% pruning, the test accuracy increased from 97.35% to 98.30%, indicating that pruning helped remove redundant parameters and possibly reduced overfitting. As shown in Figure 2, accuracy began to degrade gracefully at higher

pruning rates. Even at 40% pruning, the model retained acceptable performance, demonstrating its robustness to parameter reduction.

IX. Android Interface

Figure 3 shows the Android application's user interface during real-time intrusion detection. It displays benign and malicious detection results along with the monitoring interface.

(a) Initial State (b) Benign Traffic

(c) Malicious Traffic

Fig. 3: (a) Initial State, (b) Detection of benign traffic, (c) Detection of malicious traffic

X. Conclusions

This work presents a lightweight, real-time Intrusion Detection System (IDS) designed for Android mobile platforms using a fine-tuned version of the TSANet deep learning model. The system monitors network traffic using Android's `VpnService`, extracts a feasible subset of flow features, and classifies traffic as benign or malicious.

Due to the security and permission limitations of Android's `VpnService` API, only a limited subset of features can be extracted in real time on mobile devices. To compensate for the missing features, appropriate default values and interpolation mechanisms were employed, and the model was fine-tuned accordingly to maintain performance under these constraints.

Experimental results demonstrate high accuracy (93.2%) even with the reduced feature set, and the model achieves efficient inference on mobile devices, with an average processing time of 38 ms per flow.

Our solution is deployable on non-rooted Android devices and supports real-time alerts along with SDN-based threat mitigation, making it practical for mobile environments with limited resources.

Future work:

- Improve estimation techniques for missing features using lightweight regression models.
- Expand model capabilities to detect zero-day attacks using anomaly detection.
- Integrate federated learning for continuous model improvement without compromising user data privacy.
- Extend system support for encrypted traffic using advanced packet metadata analysis.

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